# Computer Vision Winter 2026 - Exercise 3

Machine Learning for Computer Vision - TU Dresden

November 9 – 13, 2026

## Bilateral Filters

Recall the definition of a bilateral filter given in Definition 14 of the lecture:

$$\beta(f)(v) = \frac{1}{\nu(f)(v)} \sum_{v' \in N(v)} \omega_s(d_s(v, v'))\omega_c(d_c(f(v), f(v')))f(v')$$

1. Implement a function which calculates the results of applying a bilateral filter to an image. The output of a bilateral filter is determined by the choice of $d_s, d_c, \omega_s, \omega_c$ and $N$. We want to be able to observe the results of different choices of $d_s, d_c, \omega_s\, \omega_c$ and $N$. Therefore the function which implements the bilateral filter should receive the following arguments:
   - a function $d_s$, which computes a spatial distance metric between two pixels $v, v'$.
   - a function $d_c$, which computes a color distance metric between the color values of two pixels $f(v), f(v')$.
   - a function $\omega_s$, which assigns to each resulting spatial distance metric $d_s(v, v')$ a weight between 0 and 1.
   - a function $\omega_c$, which assigns to each resulting color distance metrix $d_c(f(v), f(v'))$ a weight between 0 and 1.
   - a function $N$, which returns for each pixel $v$ a spatial neighborhood of $v$.
2. Implement the choices for $d_s, d_c, \omega_s, \omega_c$ and $N$ as presented in the lecture. Apply your bilateral filter to the `fruits_noisy.jpg` image.
3. Choose a Gaussian for $\omega_c$ and compare the results.
4. Iteratively apply bilateral filtering to an image. Compare the results to iteratively applying a Gauss filter to an image.

## Harris Corner Detection

Consider the set $V$ of pixels and the digital image $f\colon V \to \mathbb{R}$ depicted below. Moreover, consider the pixel $v$ (marked in gray) in the center of the image $f$, the neighborhood $N(v)$ including all pixels outlined in red and the function $k_v\colon N(v) \to \mathbb{R}$ such that $\forall w \in N(v) : k_v(w) := 1 \,/\, |N(v)|$.

| | | | | |
|---|---|---|---|---|
| 2 | 2 | 8 | 8 | 8 |
| 2 | 2 | 8 | 8 | 8 |
| 2 | 2 | 8 | 8 | 8 |
| 2 | 2 | 2 | 2 | 2 |
| 2 | 2 | 2 | 2 | 2 |

$f_a$

| | | | | |
|---|---|---|---|---|
| 8 | 8 | 8 | 8 | 2 |
| 8 | 8 | 8 | 2 | 2 |
| 8 | 8 | 2 | 2 | 2 |
| 8 | 2 | 2 | 2 | 2 |
| 2 | 2 | 2 | 2 | 2 |

$f_b$

5. Calculate the discrete derivatives $\partial_0 f$ and $\partial_1 f$ of the images $f_a$ and $f_b$ by using the discrete derivative filters $d_0$ and $d_1$ as seen in the lecture.
6. Calculate the structure tensor $S_k(f)$ at the marked pixel $v$ for both images $f_a$ and $f_b$.

7. Calculate the eigenvalues of the structure tensor and explain, based on these eigenvalues, whether the pixel $v$ is at a constant region, an edge or a corner in the image.

We will now implement the Harris Corner Detection algorithm.

8. For the input image `A4.png` calculate the structure tensor at each pixel.

9. Analytically derive the general formula for the eigenvalues $\lambda_1, \lambda_2$ of the structure tensor $S_k(f)$. Using your derived formula calculate the eigenvalues of the structure tensor and use them to detect all corners in the image accordingly.

10. Besides the Harris Corner Detector other means of detecting corners in an image exist. Can you come up with filters specifically designed to detect each of the corners in our input image?